

Aazain

AI Engineer - LLMs, RAG & Applied Machine Learning

+44 7721939248 | aazainumrani@gmail.com | [linkedin.com/in/aazain-umrani](https://www.linkedin.com/in/aazain-umrani) | github.com/AazainUmrani

PROFESSIONAL EXPERIENCE

Intersys Ltd | *AI Engineer*

Oct 2024 - Jul 2025

Building and deploying production LLM and ML systems

- Cut information-retrieval latency and improved response relevance by rebuilding a production RAG pipeline around vector embeddings and FastAPI-served model inference.
- Prevented output-quality regressions from reaching production by designing an LLM evaluation framework for deployed agentic AI systems (LangChain, LangGraph) on Azure cloud.
- Eliminated manual, error-prone releases by building CI/CD pipelines (Docker, GitHub Actions), enabling repeatable, tested deployment of Python-based AI services.
- Removed manual data handling from production workflows by building API integrations and microservices in Python (FastAPI) and TypeScript (NestJS).

Freelance / Independent | *AI Engineer & Consultant*

May 2022 - Dec 2023

Fine-tuning and deploying ML/NLP/CV models for 50+ clients

- Improved accuracy for niche client use cases by fine-tuning deep learning and language models with transfer learning and prompt engineering, supporting AI product development and earning 40+ five-star reviews across 50+ delivered projects.
- Delivered production-ready computer vision and NLP solutions using Python, PyTorch, and TensorFlow, working directly with clients to turn ambiguous briefs into working, evaluated models and supporting e-commerce product launching where required.

Moaice | *Software Engineer (Startup)*

Jul 2022 - Sep 2022

Shipping backend and product features for a fast-moving iOS messaging startup

- Closed core product gaps by shipping backend and product features in a small, fast-paced, cross-functional team.

PROJECTS

Production RAG Platform

- Reduced end-to-end query latency by building an optimized RAG retrieval and inference pipeline with Python, embeddings, Qdrant vector search, and FastAPI; containerized it with Docker and published it as open-source software.

MSc Thesis: Vision-Language Models vs. Object Detectors

- Improved few-shot defect-classification accuracy by adapting the Qwen2.5-VL vision-language model for a specialized visual domain, benchmarking it against YOLOv12 and RT-DETR on a multi-GPU cluster through a reproducible evaluation pipeline.

Real-Time Object Detection & Tracking

- Cut inference time to under 50ms per frame by optimising a YOLO11m-based detection and tracking pipeline for edge hardware, reaching mAP@0.5 $\approx$ 0.83 on aerial imagery.

BirdCLEF24 Kaggle Competition

- Placed in the top 200 participants globally by iterating a CNN-based bioacoustic classifier against a rigorous, metrics-driven evaluation loop.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, C/C++

LLM & GenAI: LangChain, LangGraph, RAG, Prompt Engineering, Fine-Tuning, Agentic AI Systems, Vector Databases, Large Language Models (LLMs), Vector Search

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, NLP, Computer Vision, Transformer Models, Vision-Language Models

MLOps & Cloud: Docker, CI/CD (GitHub Actions), Azure, AWS, GCP, Model Serving, Distributed Compute (HPC / Slurm)

Backend: FastAPI, NestJS, REST APIs, Git, API Integration

Product & Client Delivery: E-commerce Product Launching, B2B SaaS, AI Product Development, Dashboard Development

EDUCATION

Queen's University, Belfast UK

Sep 2025 - Aug 2026

MSc, Artificial Intelligence

Belfast, UK

- **Achievements:** Distinction (first semester)

- **Coursework:** Machine Learning, Natural Language Processing, Computer Vision, AI for Cyber Security, Knowledge Engineering

NUST, Islamabad, Pakistan

Nov 2020 - Jun 2024

BE, Computer Engineering (GPA: 3.15/4.0)

Islamabad, Pakistan

- **Achievements:** Merit Certificate for Distinguished Student